Primary Visual Pathways (II)

Introduction to Computational and Biological Vision

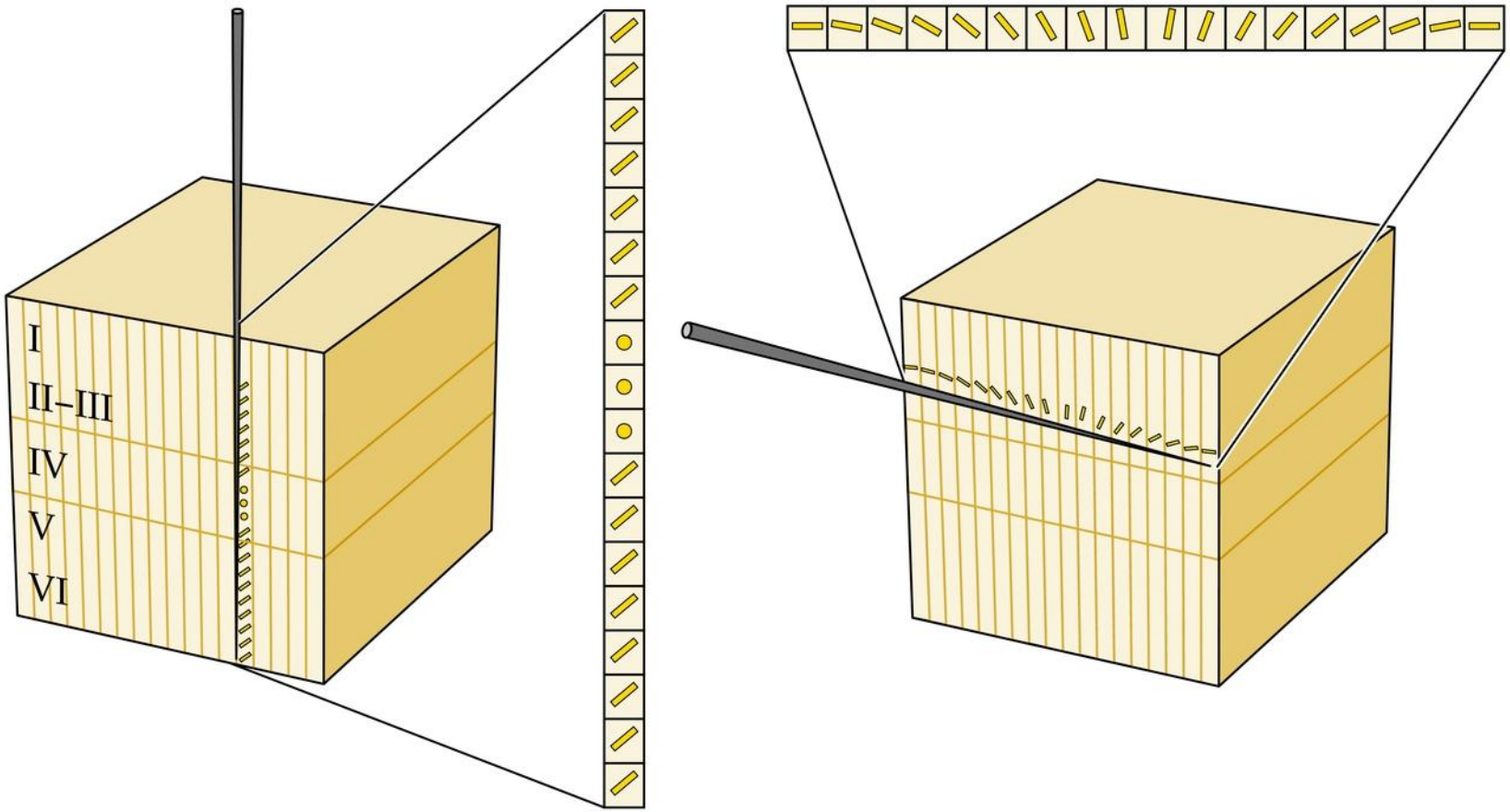
CS 202-1-5261

Computer Science Department, BGU

Ohad Ben-Shahar

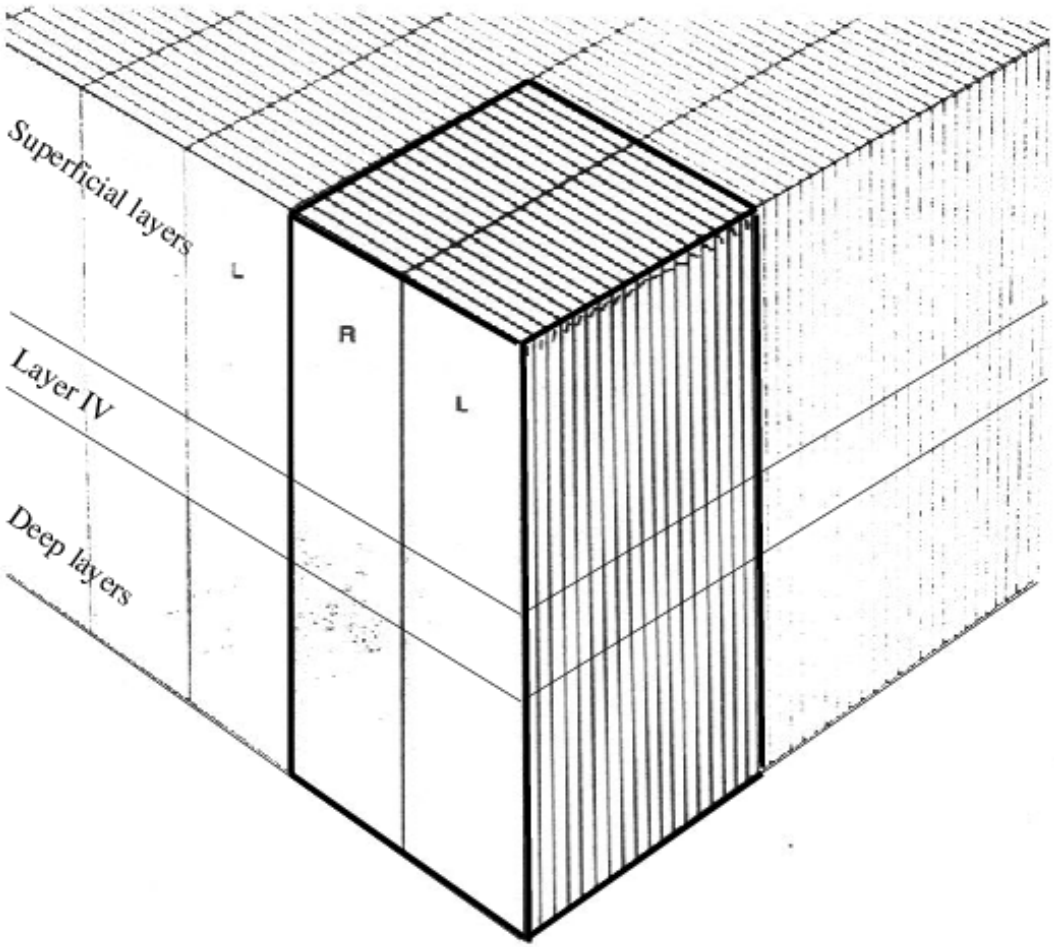
Primary visual pathway

Orientation columns



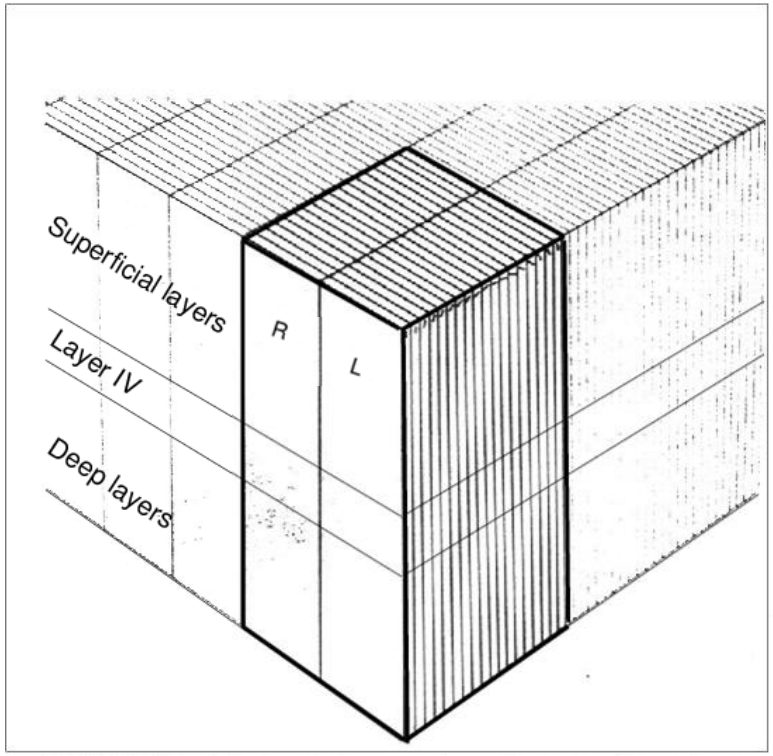
Primary visual pathway

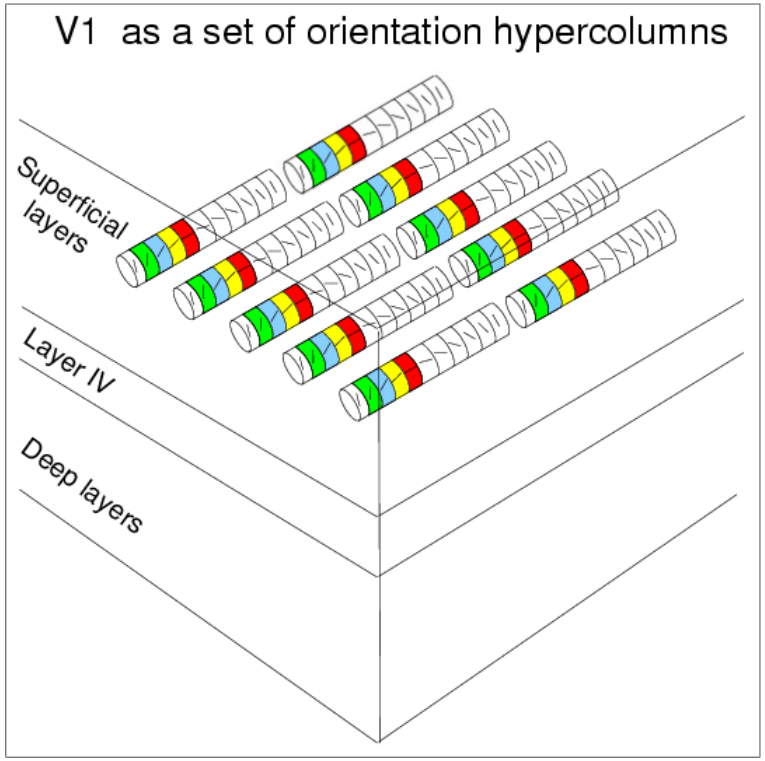
Ice cube model



Primary visual pathway

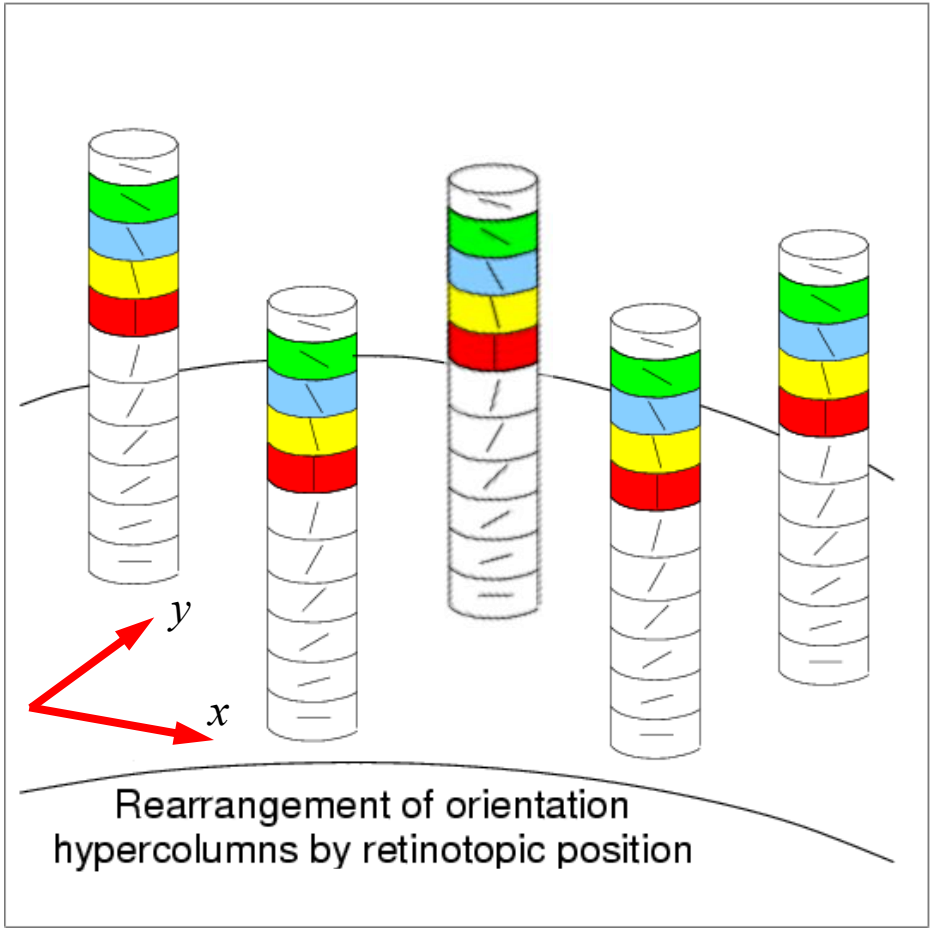
Abstraction: from primary visual cortex to geometry





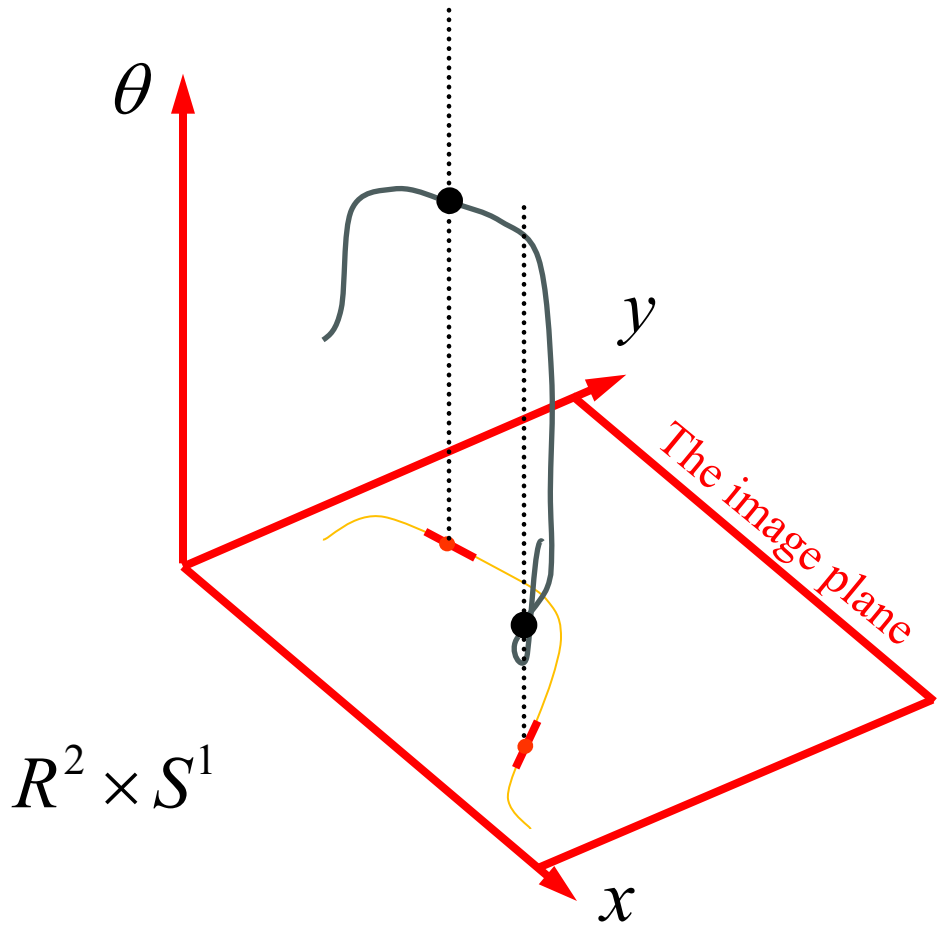
Primary visual pathway

Abstraction: from primary visual cortex to geometry



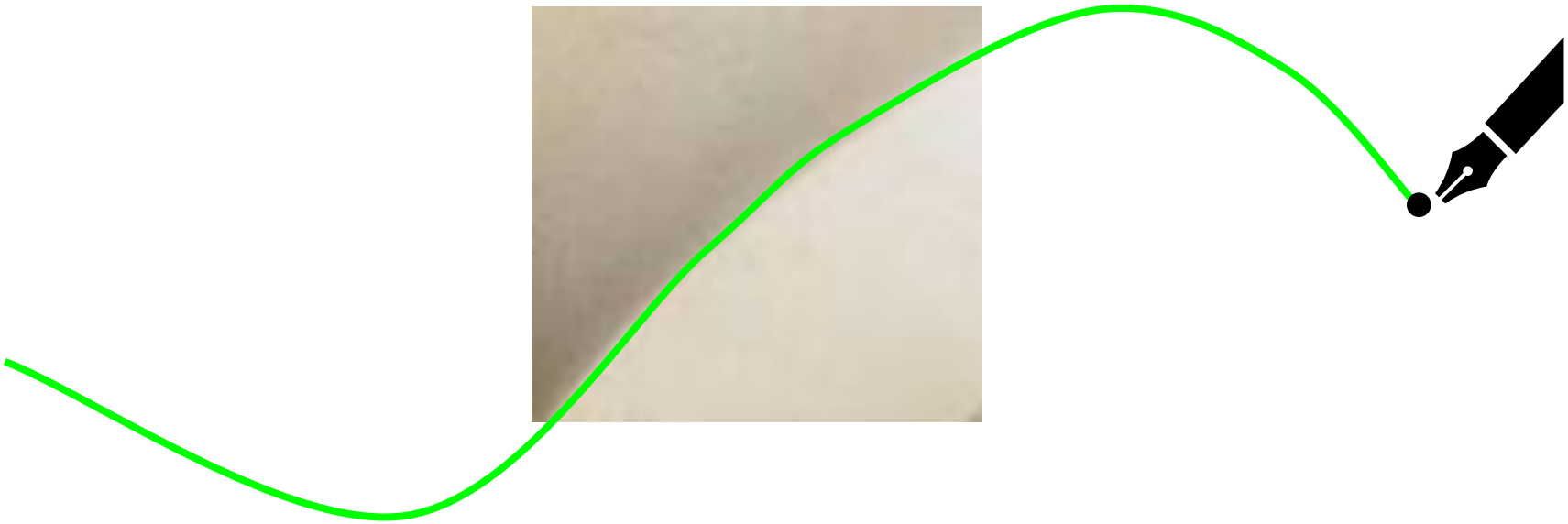
Primary visual pathway

Abstraction: from primary visual cortex to geometry



Primary visual pathway

2D Curves and orientation



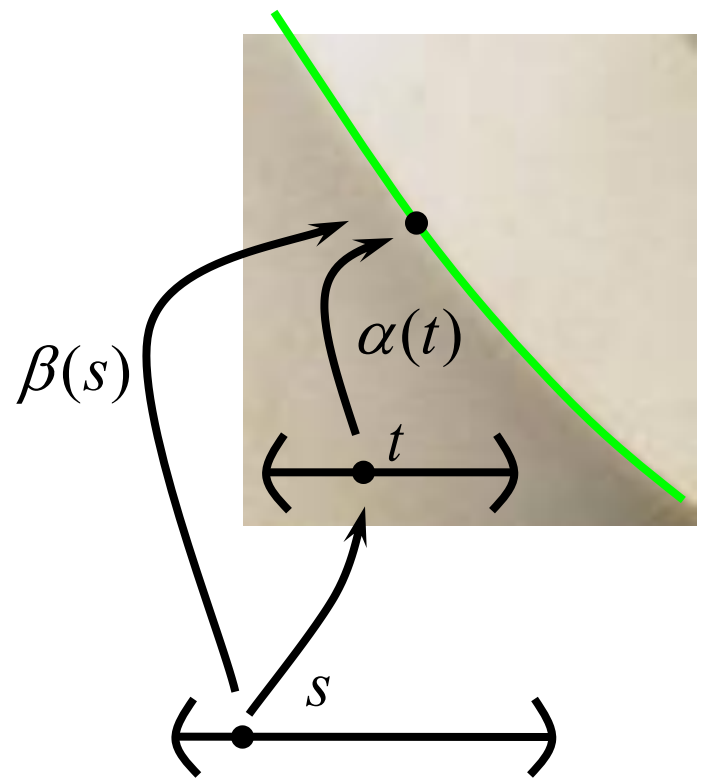
Primary visual pathway

2D Curves and orientation

Definition: A parametrized differentiable 2D **curve** is a *differentiable* map of an open interval $A = (a, b)$ into R^2

$$\alpha: (a, b) \to R^2$$
$$\alpha(t) = (x(t), y(t))$$

Definition: The set of points $\alpha(A)$ is called the **trace** of the curve α.



Definition: Let $\alpha(t)$ be a curve and let $h(s)$ be a differentiable function on the open interval J

$$h: J \to (a, b)$$

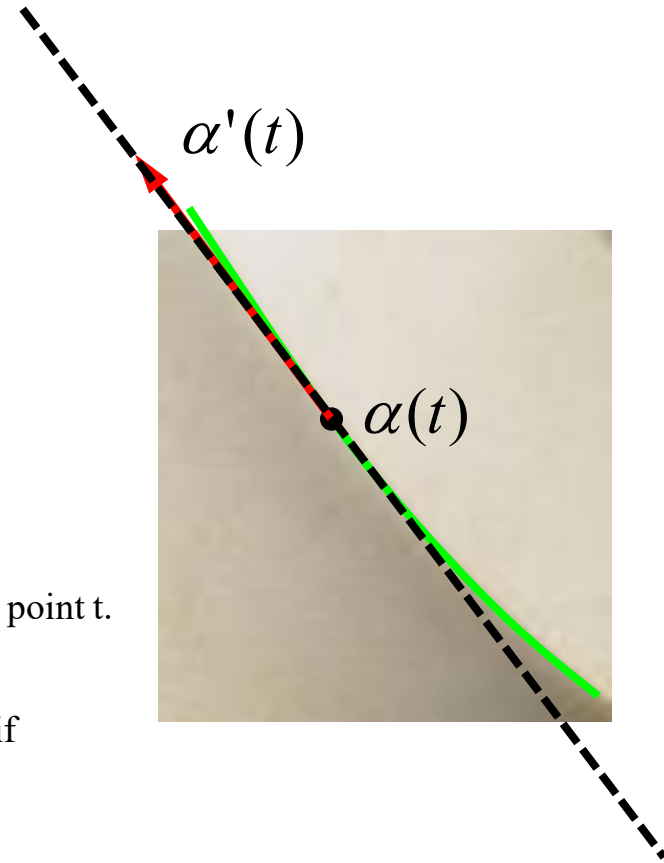
Then the composite function

$$\beta(s) = \alpha(h(s)): J \to R^2$$

is also a *curve* and is called a **reparameterization** of α by h.

Primary visual pathway

2D Curves and orientation



Definition: The derivative vector

$$\alpha'(t) = (x'(t), y'(t))$$

is called the **velocity vector** of α. |α'| is called the **speed** of the curve at t.

$$|\alpha'(t)| = \sqrt{(x'(t))^2 + (y'(t))^2}$$

Definition: The line curve $\alpha(t) + s \cdot \alpha'(t)$ is called the **tangent line** of α at point t.

Definition: A parametrized differentiable 2D curve α(t) is a said to be **regular** if

$$\alpha'(t) \neq 0 \quad \forall t \in (a, b)$$

Primary visual pathway

2D Curves and orientation

Example

$\alpha_1(t) = (t^3 - 4t, t^2 - 4)$ is a parameterized differentiable curve!

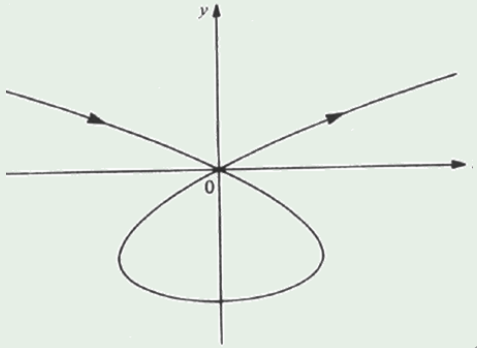
It's velocity vector is $\alpha_1(t) = (3t^2 - 4, 2t)$

It's speed at time t is $|\alpha_1'(t)| = \sqrt{(3t^2 - 4)^2 + (2t)^2} = \sqrt{9t^4 - 20t^2 + 16}$

Is it regular? YES!

- Approach 1: $\exists t\ |\alpha_1'(t)| = 0$? (i.e., does it ever "stop"?)
- Approach 2: $\exists t\ \alpha_1(t) = 0$?

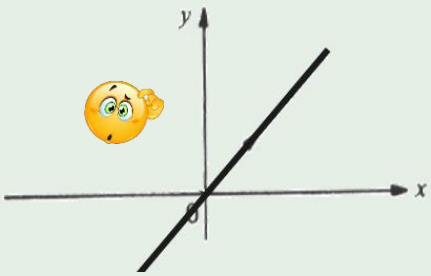
Easier to observe that this cannot happen



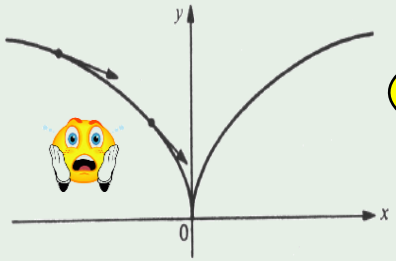
Primary visual pathway

2D Curves and orientation

Example – Are they regular?



$\alpha_2(t) = (t^3, t^3)$

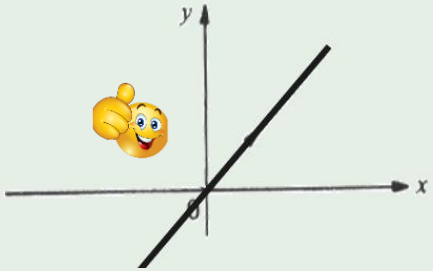


$\alpha_3(t) = (t^3, t^2)$

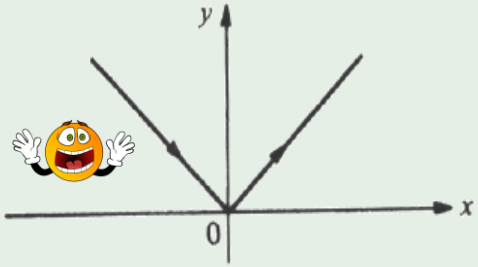
Rules of thumb:

A kink in the trace = no regular curve exist

No kink = both behaviors are possible



$\alpha_2(t) = (t, t)$



$\alpha_4(t) = (t, |t|)$

Primary visual pathway

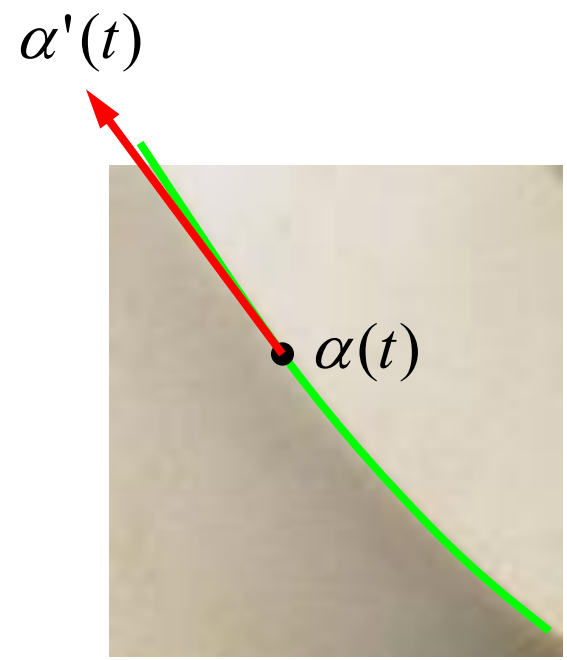
2D Curves and orientation

Definition: The **arclength** of a regular curve from the point t_0 is

$$s(t) = \int_{t_0}^{t} |\alpha'(\tau)| d\tau$$

Definition: A curve $\alpha(t)$ with $|\alpha'(t)| = 1$ is

- called **unit speed** curve
- said to be given in **arclength parametrization**,
- said to or to be **arclength parameterized.**



Theorem: If α is a regular curve in R^2, then there exists a unique reparameterization β of α such that β has unit speed.

It is always possible to reparametrize a regular curve such that its trace is "plotted" at constant unit speed!

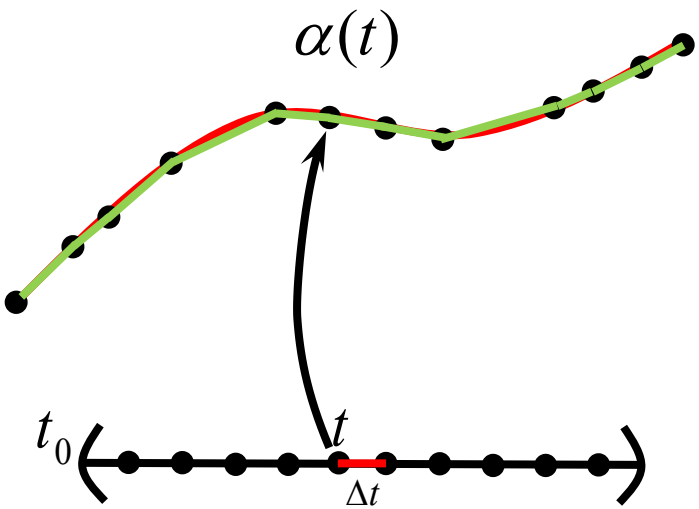
Conclusion: A unit speed parametrization of a regular piece of edge structures is a unique representation.

Primary visual pathway

Aside: Arclength

$$s_{\Delta t}(t) \cong \sum_{i=0}^{N-1} |\alpha(t_0 + (i+1)\Delta t) - \alpha(t_0 + i\Delta t)|$$

$$s_{\Delta t}(t) \cong \sum_{i=0}^{N-1} \left| \frac{\alpha(t_0 + (i+1)\Delta t) - \alpha(t_0 + i\Delta t)}{\Delta t} \right| \Delta t$$



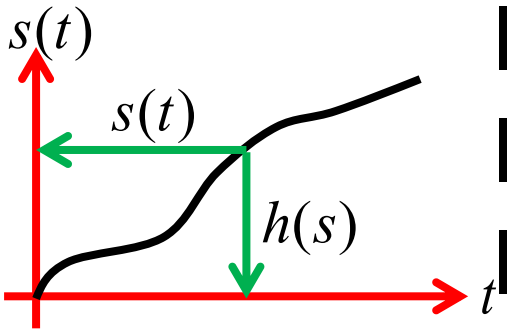
$$s(t) = \lim_{\Delta t \to 0} s_{\Delta t}(t) = \int_{t_0}^{t} |\alpha'(\tau)| d\tau$$

$$\frac{ds}{dt} = |\alpha'(t)|$$

Primary visual pathway

Aside: Arclength reparametrization of *regular* curves

$$\left.\begin{array}{l} \boldsymbol{s(t) = \int_{t_0}^{t} |\alpha'(\tau)| d\tau} \\ \boldsymbol{\alpha'(t) \neq 0} \end{array}\right\} \quad \begin{array}{l} \boldsymbol{s(t)} \text{ **is strictly monotonic!**} \\ \boldsymbol{s'(t) = |\alpha'(t)|} \end{array}$$



$$s(t):[t_0,t_1]\to[0,L] \quad \Rightarrow \quad \exists h(s):[0,L]\to[t_0,t_1] \text{ s.t. } h(s(t))=t \quad \Rightarrow \quad h=s^{-1}$$

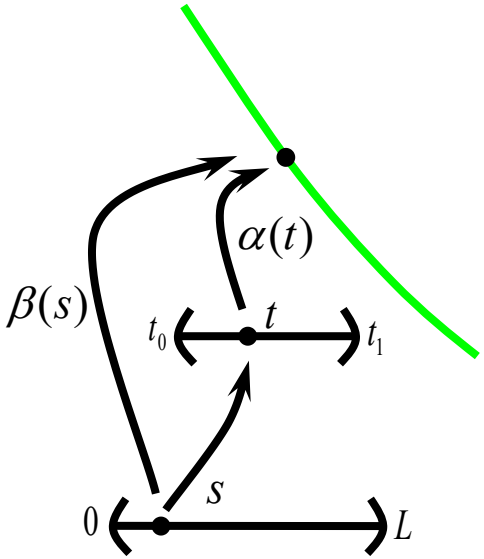
Consider now the following reparametrization of α(t)

$$\beta(s)=\alpha(\overbrace{h(s)}^{t}):[0,L]\to R^2$$

What is its speed?

$$\frac{ds}{dt}=|\alpha'(t)|$$

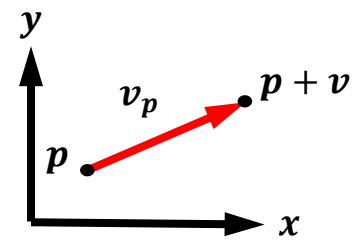
$$|\beta'(s)|=\left|\frac{d\alpha}{dt}\frac{dt}{ds}\right|=\left|\alpha'(t)\frac{1}{|\alpha'(t)|}\right|=1$$



Primary visual pathway

Not to be confused with "tangent line"

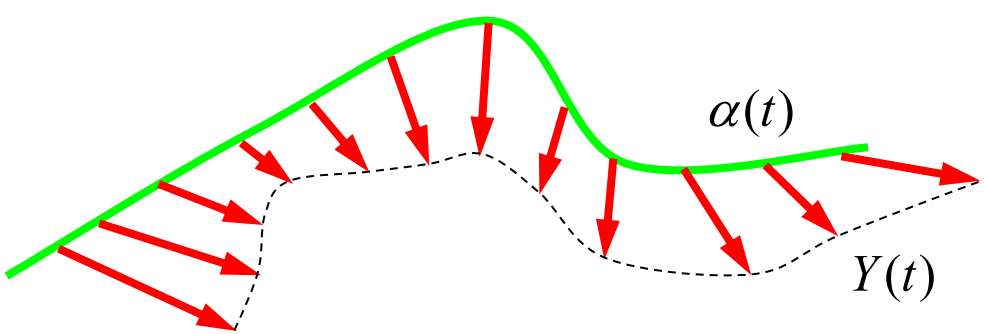
2D Curves and orientation



Definition: A **tangent vector** $\boldsymbol{v_p}$ to $\boldsymbol{R^2}$ is a pair $(\boldsymbol{v}, \boldsymbol{p})$ where $\boldsymbol{v}$ is a vector and $\boldsymbol{p}$ is its application point .

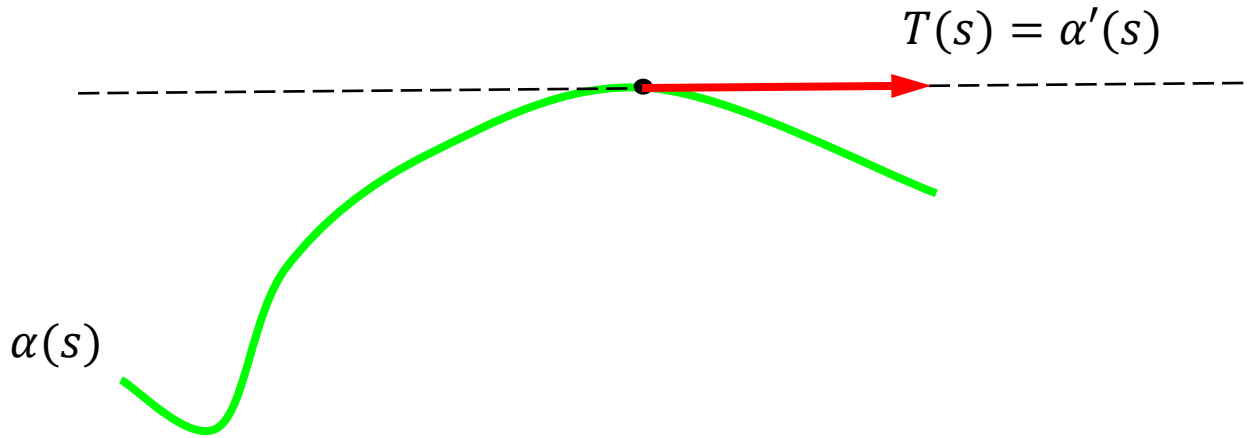
Definition: Let $\boldsymbol{p}$ be a point of R^2. The set $\boldsymbol{T_p(R^2)}$ consisting of all tangent vectors $\boldsymbol{v_p}$ is called the **tangent space** of $\boldsymbol{R^2}$ at $\boldsymbol{p}$.

Definition: A **vector field** on a curve $\boldsymbol{\alpha}: \boldsymbol{I} \to \boldsymbol{R^2}$ is a function that assigns to each $\boldsymbol{t} \in \boldsymbol{I}$ a tangent vector $Y(t)$ where $\boldsymbol{Y(t)} \in \boldsymbol{T_{\alpha(t)}(R^2)}$ **.** The vector field is said to be differentiable if $\boldsymbol{Y(t)}$ is differentiable.



Primary visual pathway

2D Curves and orientation



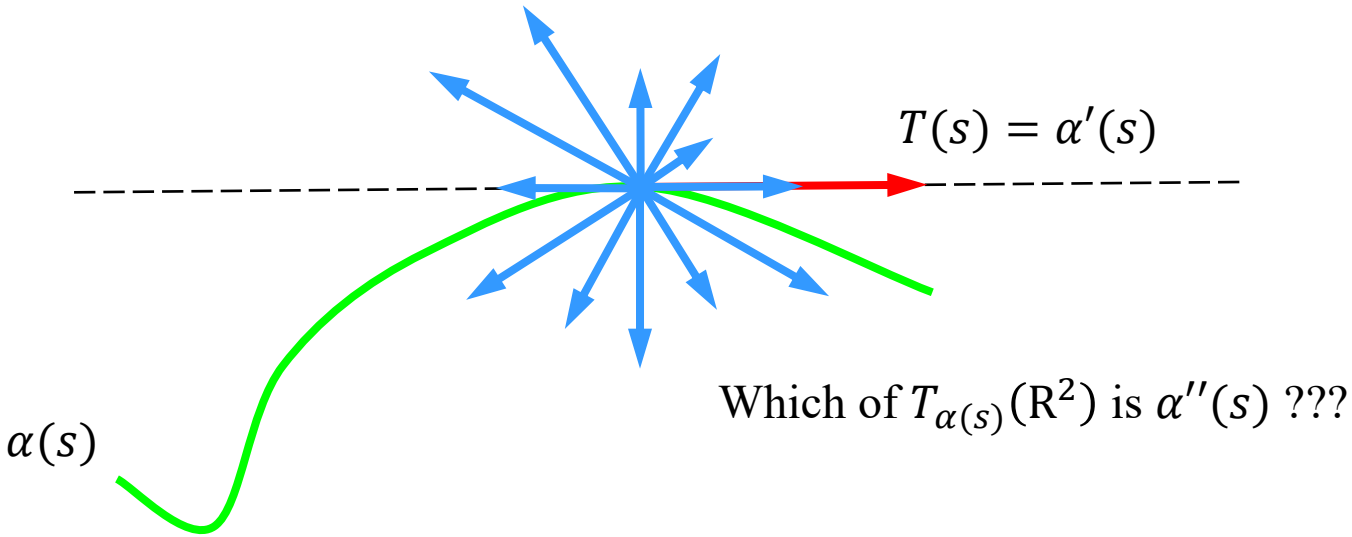
Unit speed curve : $\|\alpha'(s)\| = 1$

Unit tangent vector field : $T(s) = \alpha'(s)$

Not to be confused with "tangent line" or the general concept of a tangent vector

Primary visual pathway

2D Curves and orientation



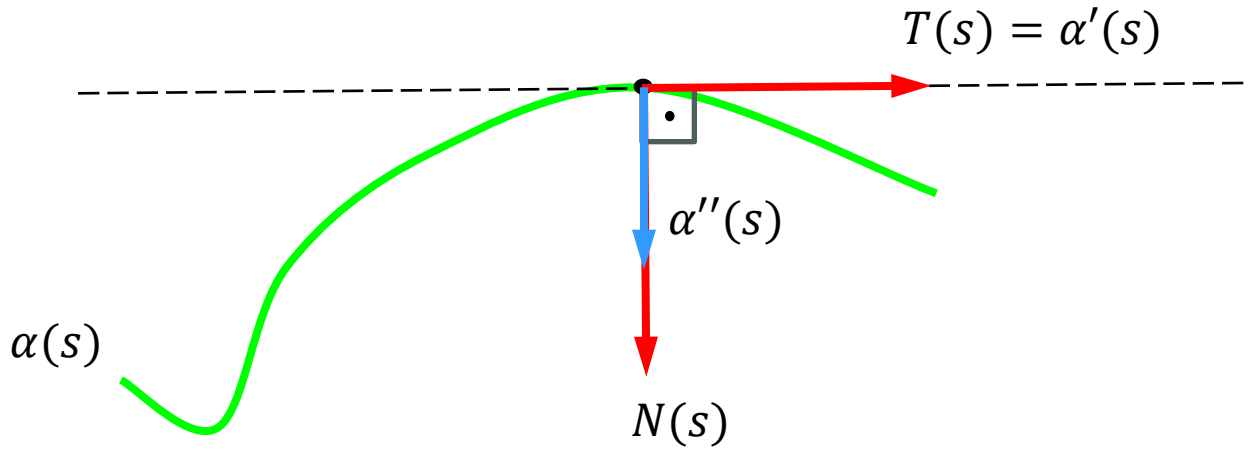
Unit speed curve : $\|\alpha'(s)\| = 1$

Unit tangent vector field : $T(s) = \alpha'(s)$

acceleration vector : $T'(s) = \alpha''(s) = \big(x''(s), y''(s)\big) \in T_{\alpha(s)}(R^2)$

Primary visual pathway

2D Curves and orientation



Unit speed curve : $\|\alpha'(s)\| = 1$

Unit tangent vector field : $T(s) = \alpha'(s)$

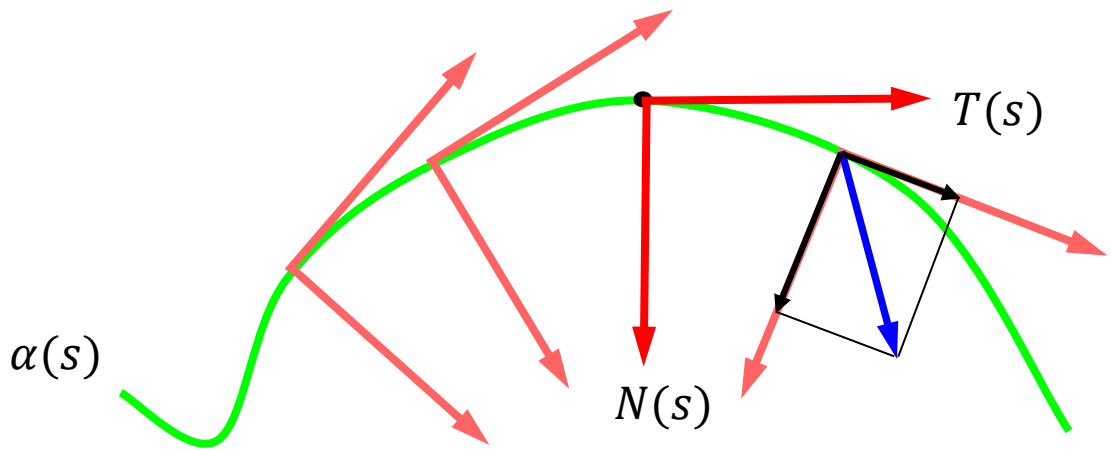
The Normal vector

The curve's **Curvature**

Curvature vector : $T'(s) = \alpha''(s) = (x''(s), y''(s)) = \|T'(s)\| N(s) = \boldsymbol{\kappa(s)} N(s)$

Primary visual pathway

2D Curves and orientation



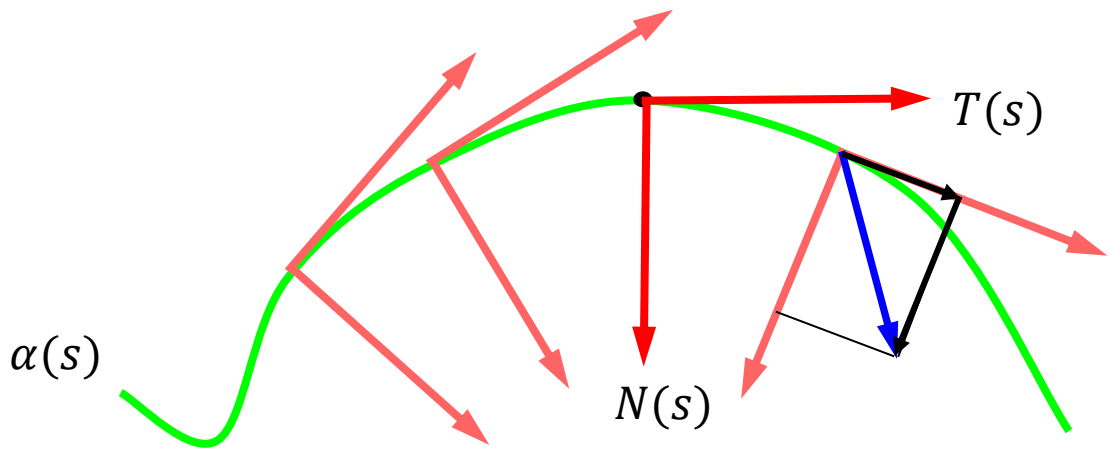
$$\begin{aligned} T \cdot T &= 1 \\ N \cdot N &= 1 \\ T \cdot N &= 0 \end{aligned} \quad \Longrightarrow \quad \begin{aligned} T \cdot T' &= 0 \\ N \cdot N' &= 0 \\ T \cdot N' &= -T' \cdot N \end{aligned}$$

The principle of
orthogonal expansion

$$V = (V \cdot T)T + (V \cdot N)N$$

Primary visual pathway

2D Curves and orientation



$$T \cdot T' = 0$$
$$N \cdot N' = 0$$
$$T \cdot N' = -T' \cdot N$$

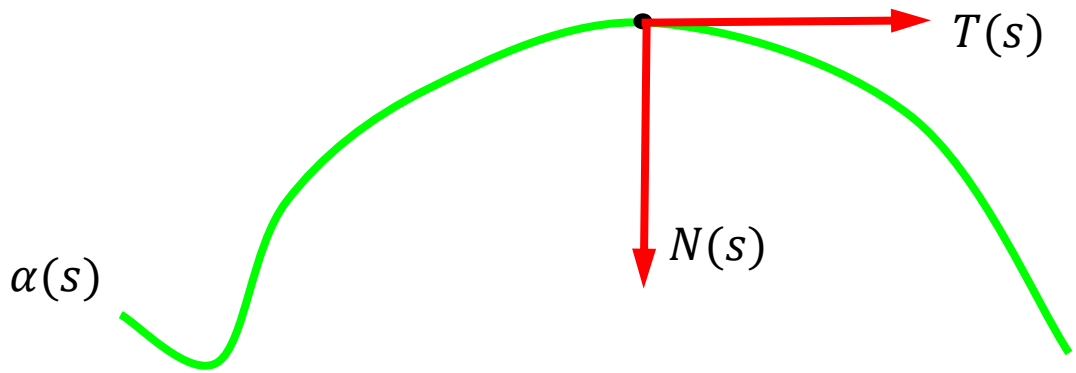
$$\Rightarrow \quad T' = (\cancel{T' \cdot T})T + \overset{\kappa}{(T' \cdot N)}N$$
$$N' = (N' \cdot T)T + (\cancel{N' \cdot N})N$$

The principle of
orthogonal expansion

$$V = (V \cdot T)T + (V \cdot N)N$$

Primary visual pathway

2D Curves and orientation

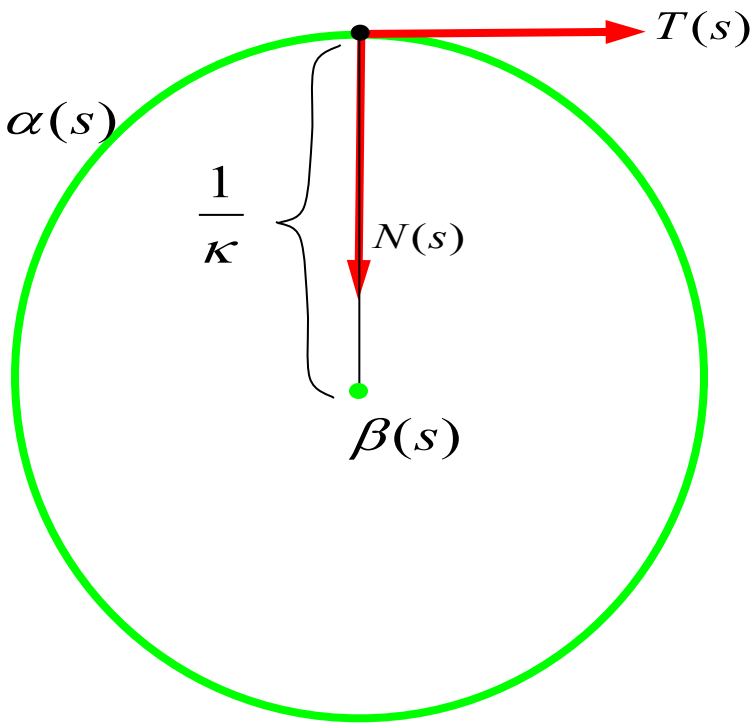


$$\begin{pmatrix} T' \\ N' \end{pmatrix} = \begin{pmatrix} 0 & \kappa \\ -\kappa & 0 \end{pmatrix} \begin{pmatrix} T \\ N \end{pmatrix}$$

The **Frenet frame** and the **Frenet equation** for plane curves

Primary visual pathway

2D Curves and orientation



Lemma: If α(s) is a unit-speed plane curve with constant curvature κ>0, then α is part of a circle of radius 1/κ

Proof: Consider the curve

$$\beta(s) = \alpha(s) + \frac{1}{\kappa} N(s)$$

Using the assumption and the Frenet equation for α(s) we get

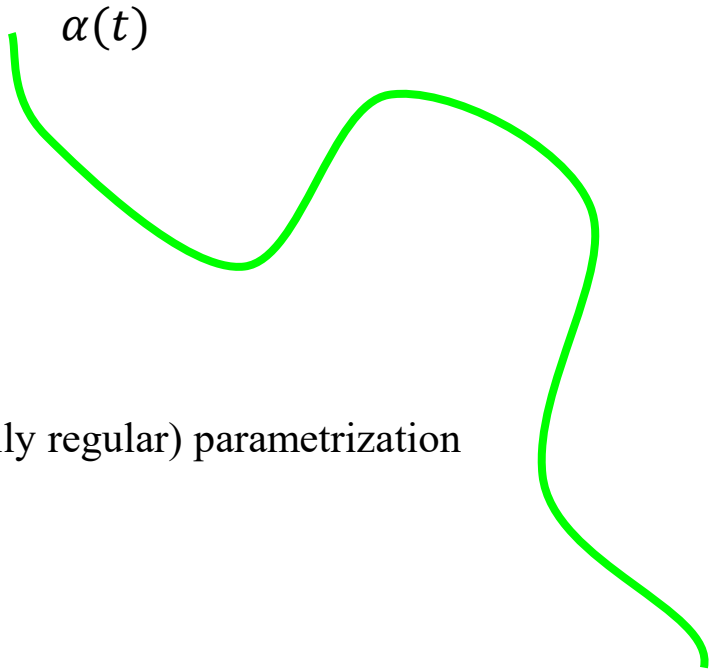
$$\beta'(s) = T(s) + \frac{1}{\kappa}(-\kappa T(s)) = 0$$

β(s) is therefore a constant curve whose single point trace has the following constant distance from α(s)

$$\|\beta(s) - \alpha(s)\| = \left\|\alpha(s) + \frac{1}{\kappa} N(s) - \alpha(s)\right\| = \left\|\frac{1}{\kappa} N(s)\right\| = \frac{1}{\kappa}$$

Primary visual pathway

2D Curves and orientation



Curvature of a regular curve given in general (not necessarily regular) parametrization

$$\alpha(t) = (x(t), y(t))$$

$$\kappa(t) = \frac{|x'y'' - y'x''|}{(x'^2 + y'^2)^{\frac{3}{2}}}$$

Primary visual pathway

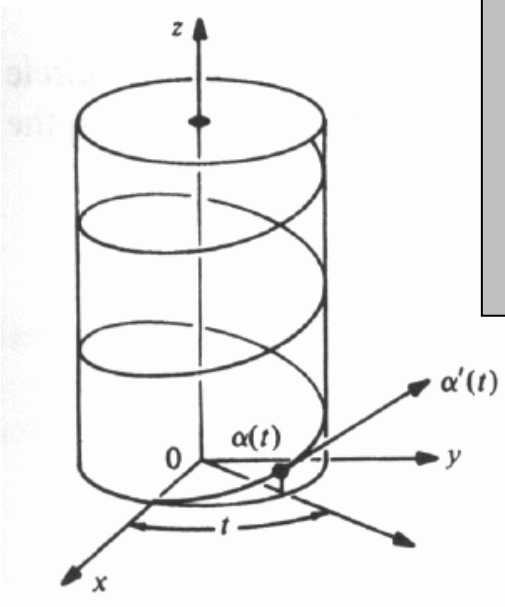
3D Curves and orientation

Space curves : $\alpha(t) = (x(t), y(t), z(t))$

Velocity vector : $\alpha'(t) = (x'(t), y'(t), z'(t))$

Speed : $|\alpha'(t)| = \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}$

Tangent line : $\alpha(t) + s \cdot \alpha'(t)$

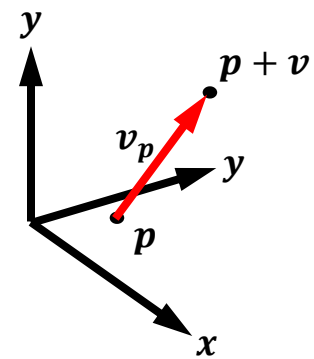


$\alpha(t) = (a\cos(t), a\sin(t), bt)$

Primary visual pathway

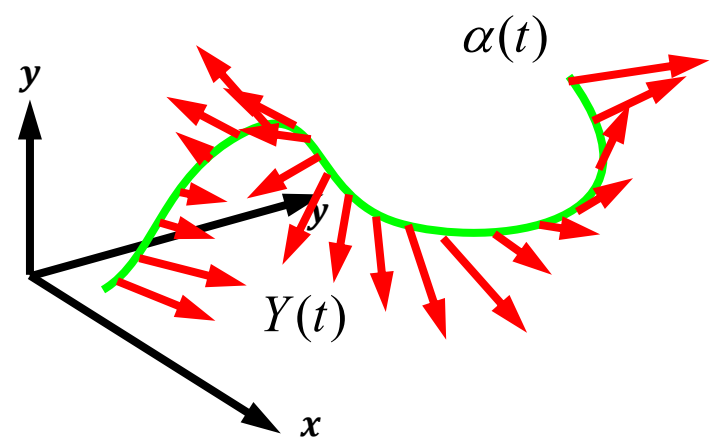
3D Curves and orientation

Definition: A **tangent vector** $\boldsymbol{v_p}$ to $\boldsymbol{R^2}$ is a pair $(\boldsymbol{v}, \boldsymbol{p})$ where $\boldsymbol{v}$ is a vector and $\boldsymbol{p}$ is its application point .



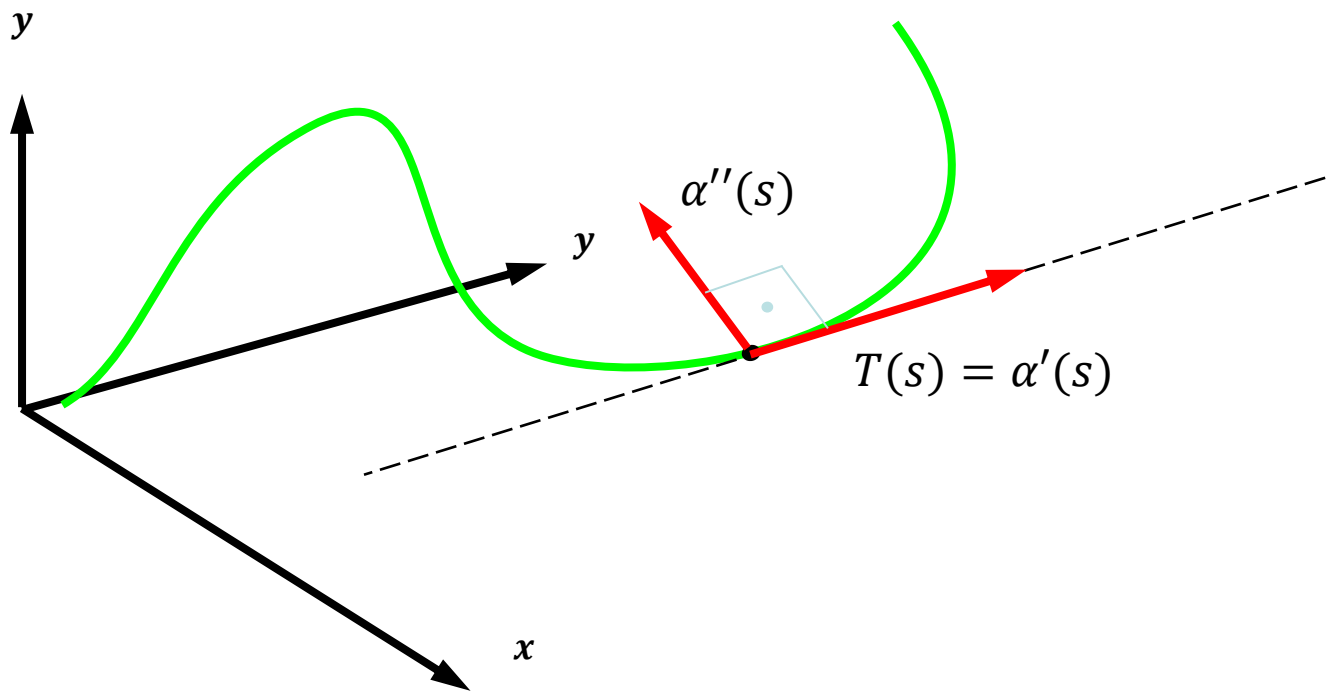
Definition: Let $\boldsymbol{p}$ be a point of 3^2**.** The set $\boldsymbol{T_p(R^{23})}$ consisting of all tangent vectors $\boldsymbol{v_p}$ is called the tangent space of $\boldsymbol{R^3}$ at $\boldsymbol{p}$.

Definition: A **vector field** on a curve $\boldsymbol{\alpha}: \boldsymbol{I} \to \boldsymbol{R^3}$ is a function that assigns to each $\boldsymbol{t} \in \boldsymbol{I}$ a tangent vector $Y(t)$ where $\boldsymbol{Y(t)} \in \boldsymbol{T_{\alpha(t)}(R^3)}$**.** The vector field is said to be differentiable if $\boldsymbol{Y(t)}$ is differentiable.



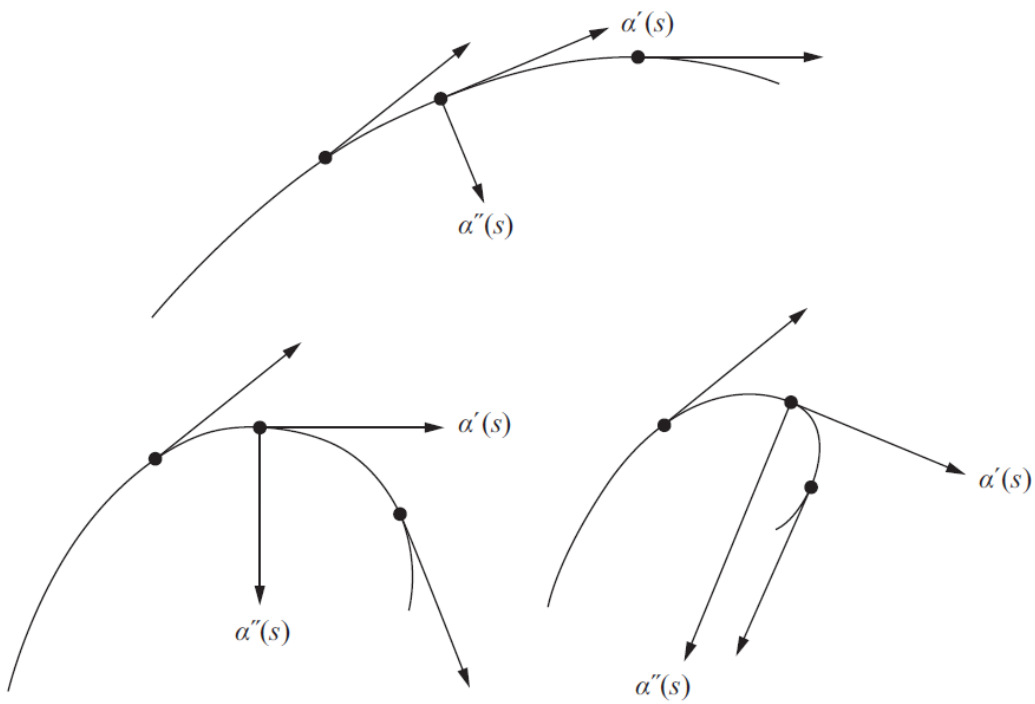
Primary visual pathway

3D Curves and orientation



Primary visual pathway

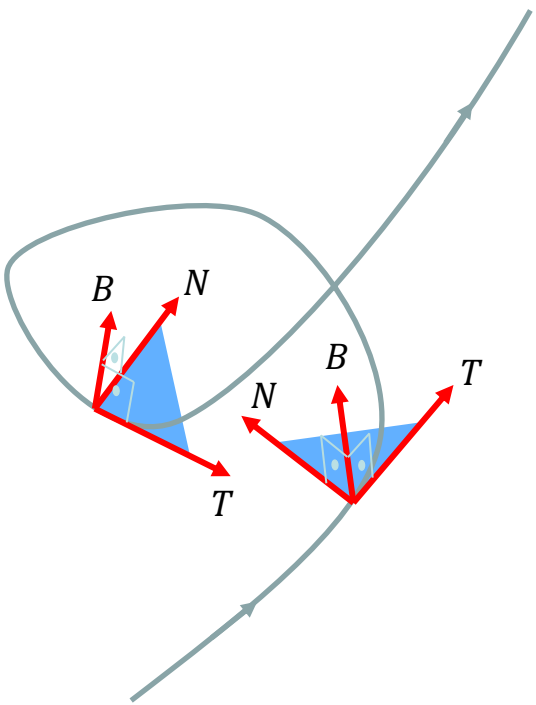
3D Curves and orientation



Definition: Let $\alpha(s): I \to R^3$ be a curve parametrized by arclength $s \in I$. $\alpha''(s)$ is called the **acceleration vector** and the number $|\alpha''(s)| = \kappa(s)$ is called the **curvature** of α at s.

Primary visual pathway

3D Curves and orientation



Definition: The plane formed by the tangent and normal vectors is called the *osculating plane*

Note that : If $\alpha(s)$ is a *planar curve* in 3D space, its osculating plane is coincident with the plane of the curve.

Note that: In regular curves the tangent vector is always defined, but the normal vector (and therefore the osculating plane) may not be.

Definition: Points $s \in A$ such that $\alpha''(s) = 0$ and $\kappa(s) = 0$ are called singular points of order 1, or *inflection points*

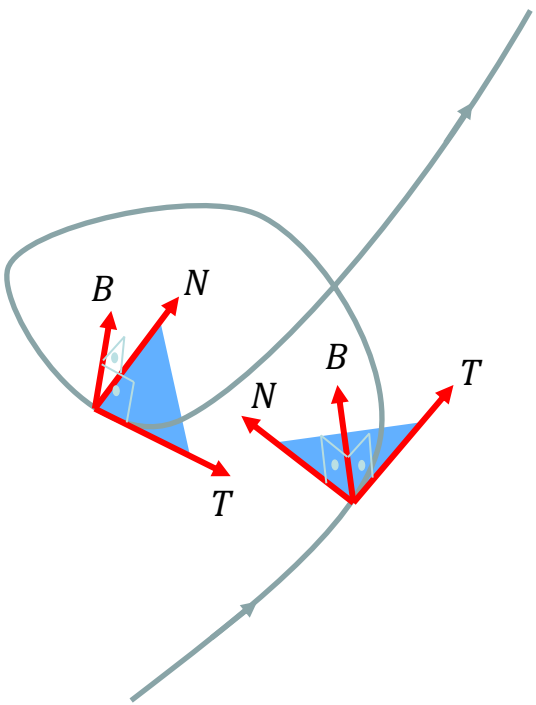
Definition: The vector $B(s) = T(s) \times N(s)$ is called the binormal vector at point $\alpha(s)$.

Studying $B'(s)$ will be central to understanding the geometry of the curve

Note that: $B(s)$ is unit vector that is always perpendicular to the osculating plane. $|B'(s)|$ thus measures the rate of change of the osculating plane as it varies with s, *i.e.* how rapidly the curves "pulls away" from its osculating plane at s.

Primary visual pathway

3D Curves and orientation



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$$\frac{\partial}{\partial s}$$

$$B'(s) = \cancel{T'(s) \times N(s)} + T(s) \times N'(s) \ = T(s) \times N'(s)$$

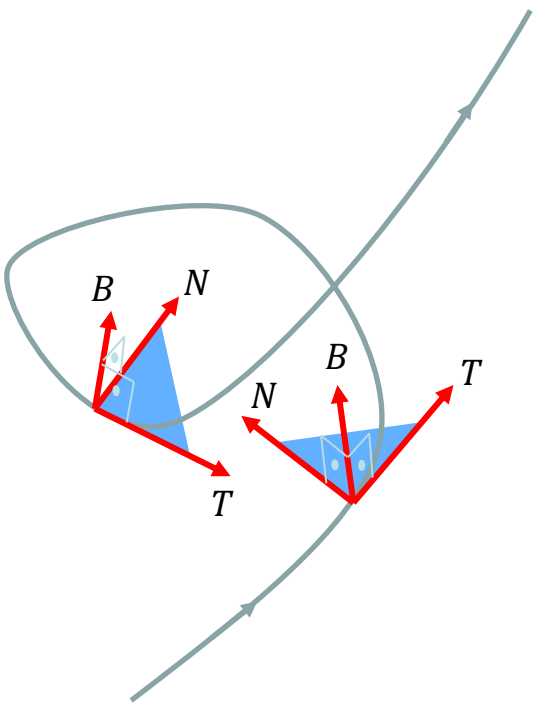
Conclusion: $B'(s)$ is orthogonal to $T(s)$, but as a derivative of a unit vector it is also orthogonal to $\mathrm{B}(s)$, hence parallel to $\mathrm{N}(s)$:

$$B'(s) = |B'(s)|N(s) \ = \tau(s)N(s)$$

So not unlike the origin of κ, let's give this size a name!

Primary visual pathway

3D Curves and orientation



torsion

Normal vector

$$B'(s) = |B'(s)|N(s) = \tau(s)N(s)$$

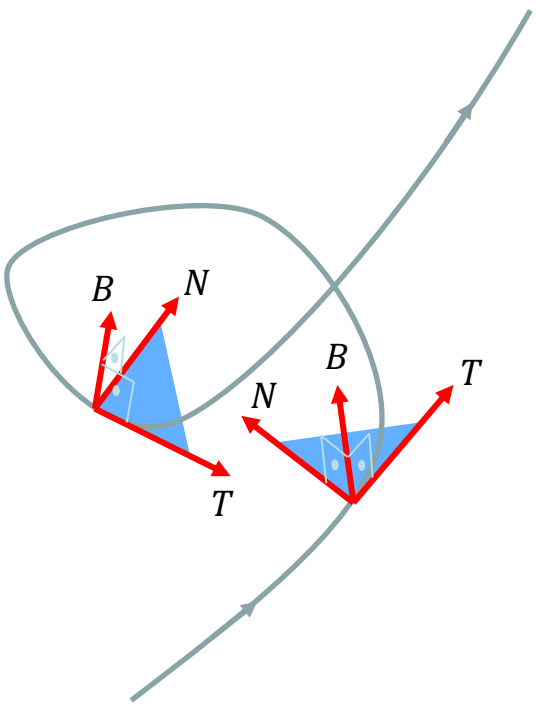
Definition: let $\alpha(s)$ be a regular space curve parametrized by arclength s **such that** $\boldsymbol{\alpha''(s) \neq 0\ \forall s}$. The number $\tau(s)$ that satisfies $B'(s) = \tau(s)N(s)$ is called the *torsion* of the curve at s

Observation: If $\alpha(s)$ is a 2D plane curve, the plane of the curve is the osculating plane at all points and thus $\tau \equiv 0$.

Definition: The three orthonormal vectors $T(s), N(s), B(s)$ are called the **Frenet Frame** of the space curve at s.

Primary visual pathway

3D Curves and orientation



Definition: The three orthonormal vectors $T(s), N(s), B(s)$ are called the **Frenet Frame** of the space curve at s.

And we've seen that:

$$T'(s) = \kappa(s)N(s)$$

$$B'(s) = \tau(s)N(s)$$

$$N'(s) = ???$$

We recall that:

$$N = B \times T$$

$$N' = B' \times T + B \times T' \quad = \tau N \times T + \kappa B \times N$$

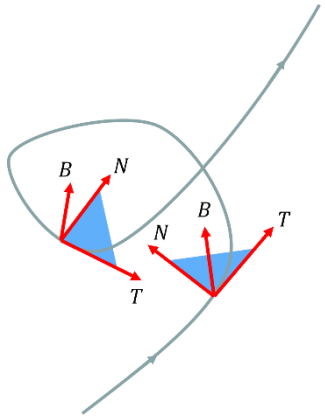
$$= -\tau T \times N - \kappa N \times B \quad = -\tau B - \kappa T$$

Because this entails…

Let's keep order of vector products consistent

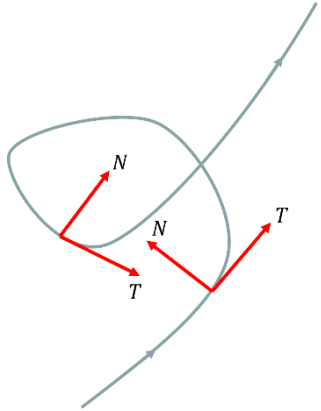
Primary visual pathway

3D Curves and orientation



The Frenet Equations for space (3D) curves:

$$\begin{bmatrix} T' \\ N' \\ B' \end{bmatrix} = \begin{bmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & -\tau \\ 0 & \tau & 0 \end{bmatrix} \begin{bmatrix} T \\ N \\ B \end{bmatrix}$$



And if the curve is planar (2D), where $\tau = 0$:

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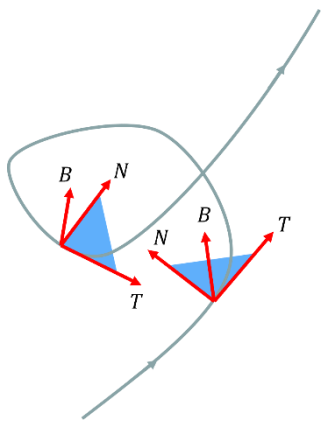
Primary visual pathway

3D Curves and orientation



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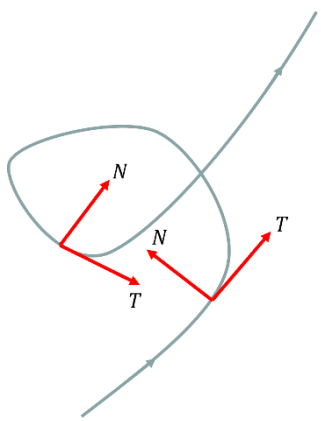
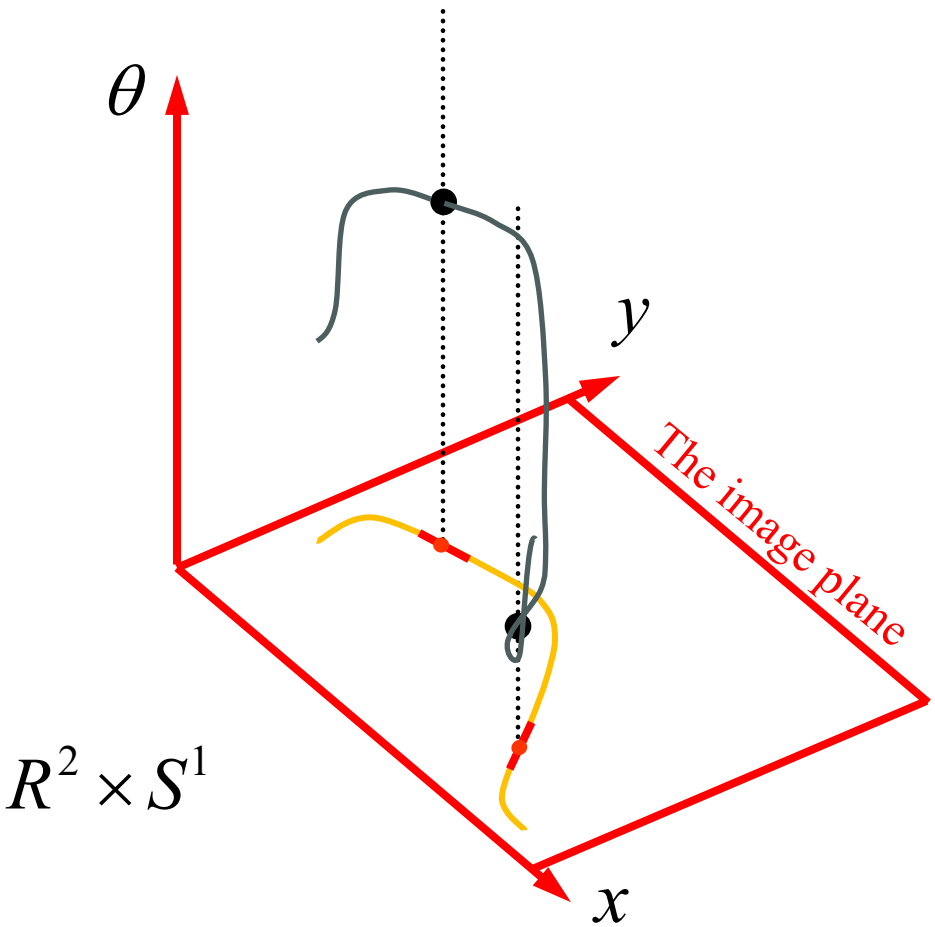


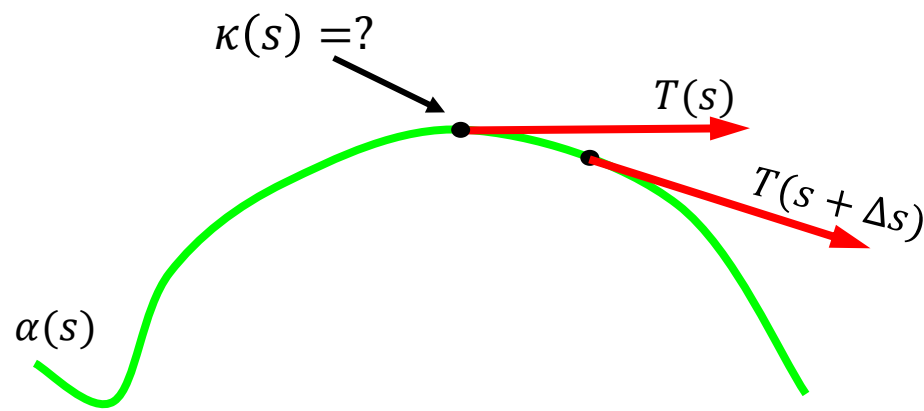
The fundamental theorem of the local geometry of curves:

Given differentiable functions $\kappa(s)>0$ and $\tau(s)$, for $s \in I$, there exists a regular parameterized curve $\alpha: I \to R^3$ such that s is $\alpha's$ arclength, $\kappa(s)$ is its curvature, and $\tau(s)$ is its torsion. Moreover, any other curve $\bar{\alpha}(s)$ that satisfies the same conditions, differs from α by a rigid transformation.

Primary visual pathway

Curvature and cortical machinery

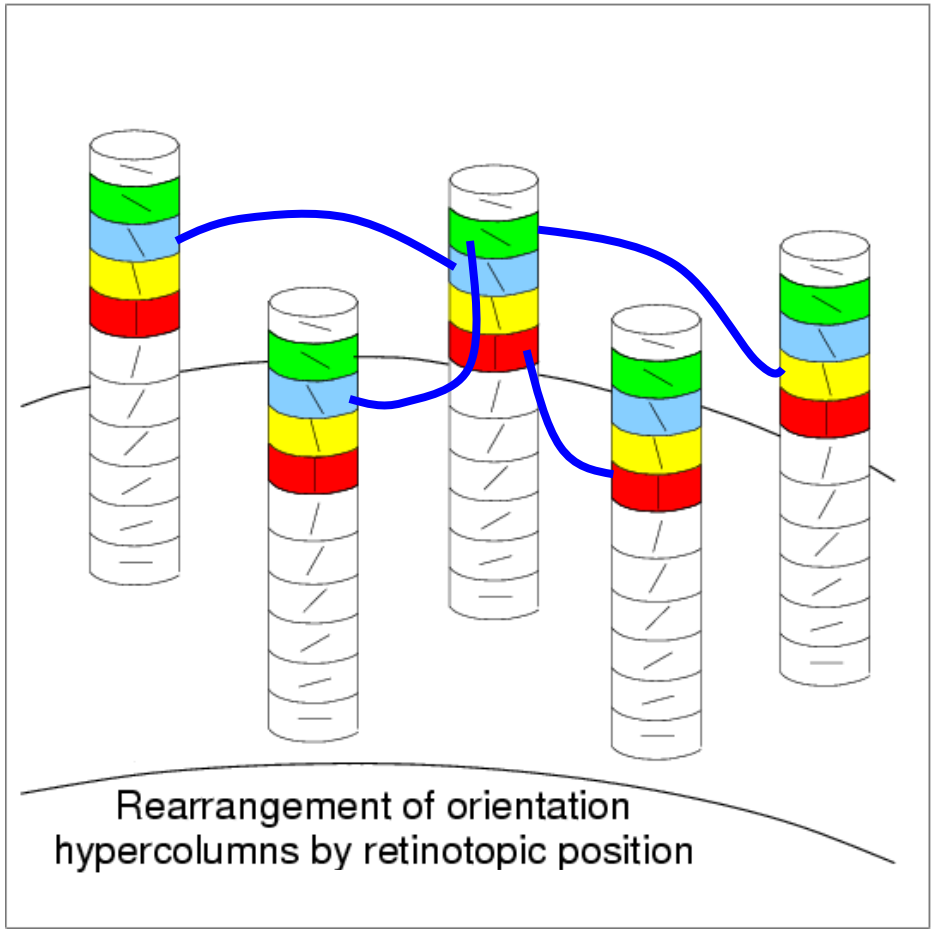


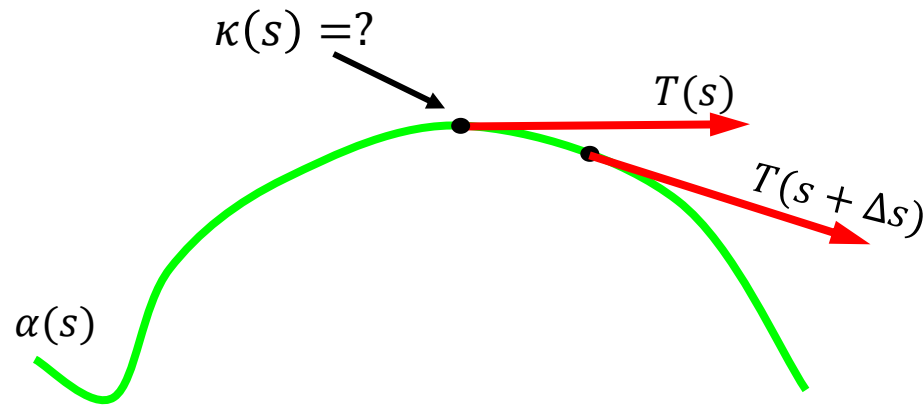


$$\kappa(s) = \|T'(s)\| = \left\| \lim_{\Delta s \to 0} \frac{\boldsymbol{T(s + \Delta s) - T(s)}}{\boldsymbol{\Delta s}} \right\|$$

Primary visual pathway

Abstraction: from primary visual cortex to geometry





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Primary visual pathway

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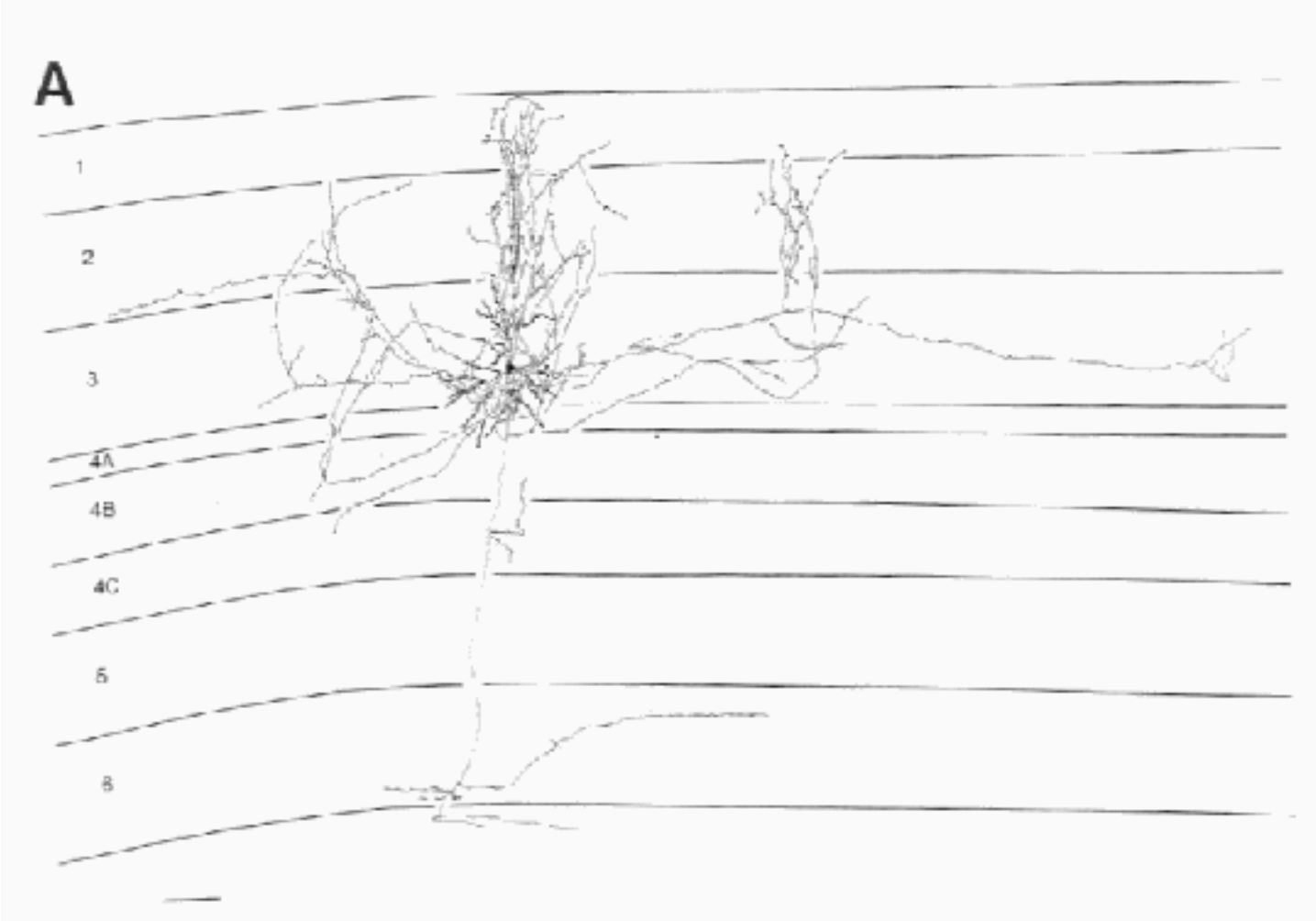
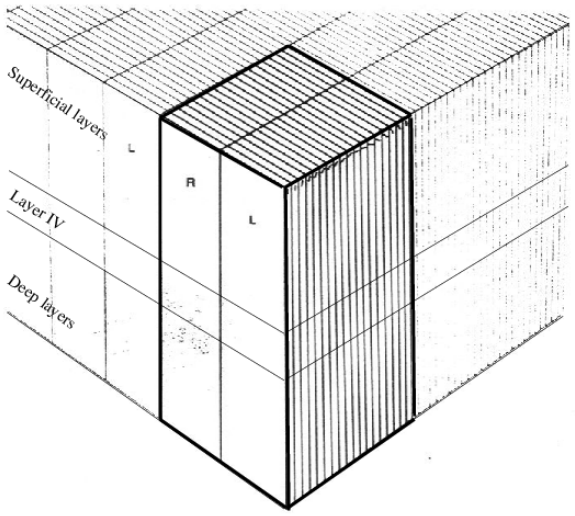
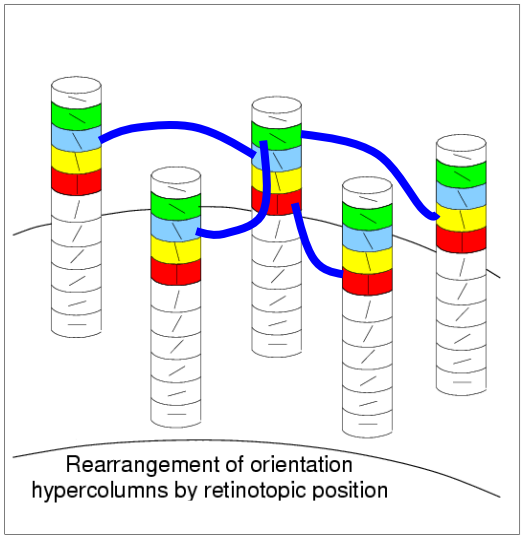
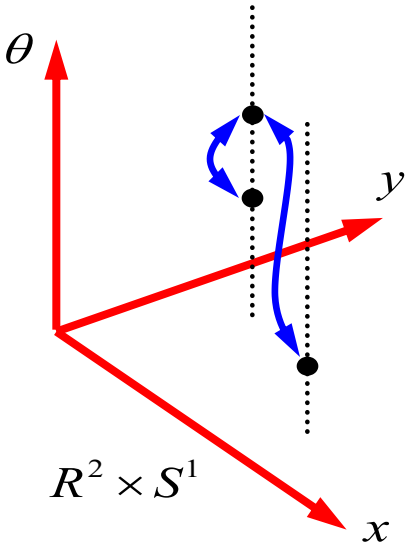


Primary visual pathway

Computing in orientation columns







What can be computed within orientation columns?
What can be computed between them?